THE FUNCTOR OF POINTS WHERE A MODULE IS FLAT IN DIMENSIONS AT LEAST N IS AN FPQC SHEAF AND LIMIT PRESERVING

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The situation defining the functor of points where the module is flat in dimensions at least n. Same chapter flat.tex, situation labelled situation-flat-dimension-n (statement, proof line 6338). Fixes what F_n is; the proof has to show the sheaf property and the limit preserving property for that functor.
- The flat locus and its behaviour under base change, and the dimension of fibres after base change. More on Morphisms, labels more-morphisms-lemma-flat-locus-base-change (proof line 6363) and more-morphisms-theorem-openness-flatness (proof lines 6399); Morphisms, label morphisms-lemma-dimension-fibre-after-base-change (proof line 6368). The first relates the non flat loci upstairs and downstairs so that flatness in dimensions at least n descends along a flat base change; openness of flatness identifies the open flat locus in the limit preserving part.
- Approximation and descent lemmas from the theory of limits. Limits, labels limits-lemma-approximate-given-relative-dimension (proof line 6402), limits-lemma-descend-finite-presentation and limits-lemma-descend-closed-immersion-finite-presentation (proof lines 6406-6407), limits-lemma-limit-dimension (proof line 6410) and limits-lemma-descend-module-flat-finite-presentation (proof line 6412). Supply a finitely presented closed subscheme of dimension below n containing the non flat locus, descend it to a stage together with its dimension bound and with the flatness of the module off it, which yields flatness in dimensions at least n at that stage.

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Source reference. Original lemma 05M6: sources/stacks-chow/tags.txt, source lines 7260–7260. Navigation pointer for \ref{limits-lemma-approximate-given-relative-dimension} used inside the counted span (line 6402); the argument that consumes it is retained in the span.

Source reference. Original lemma 01ZP: sources/stacks-chow/tags.txt, source lines 2602–2602. Navigation pointer for \ref{limits-lemma-descend-closed-immersion-finite-presentation} used inside the counted span (line 6407); the argument that consumes it is retained in the span.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Navigation pointer for \ref{limits-lemma-descend-finite-presentation} used inside the counted span (line 6406); the argument that consumes it is retained in the span.

Source reference. Original lemma 05LY: sources/stacks-chow/tags.txt, source lines 7252–7252. Navigation pointer for \ref{limits-lemma-descend-module-flat-finite-presentation} used inside the counted span (line 6412); the argument that consumes it is retained in the span.

Source reference. Original lemma 05M5: sources/stacks-chow/tags.txt, source lines 7259–7259. Navigation pointer for \ref{limits-lemma-limit-dimension} used inside the counted span (line 6410); the argument that consumes it is retained in the span.

Source reference. Original lemma 047C: sources/stacks-chow/tags.txt, source lines 5503–5503. Navigation pointer for \ref{more-morphisms-lemma-flat-locus-base-change} used inside the counted span (line 6363); the argument that consumes it is retained in the span.

Source reference. Original lemma 0399: sources/stacks-chow/tags.txt, source lines 4264–4264. Navigation pointer for \ref{more-morphisms-theorem-openness-flatness} used inside the counted span (line 6399); the argument that consumes it is retained in the span.

Source reference. Original lemma 02FY: sources/stacks-chow/tags.txt, source lines 3209–3209. Navigation pointer for \ref{morphisms-lemma-dimension-fibre-after-base-change} used inside the counted span (line 6368); the argument that consumes it is retained in the span.

Source reference. Original lemma 05MT: sources/stacks-flat/pinned-original-source.tex, source lines 6311–6311. Navigation pointer for \ref{situation-flat-dimension-n} used inside the counted span (line 6338); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 05MT.*

(1) *The functor F_n satisfies the sheaf property for the fpqc topology.*
(2) *If f is quasi-compact and locally of finite presentation and $\mathcal{F}$ is of finite presentation, then the functor F_n is limit preserving.*

Proof. Let $\{T_i \to T\}_{i \in I}$ be an fpqc covering of schemes over S. Set $X_i = X_{T_i} = X \times_S T_i$ and denote $\mathcal{F}_i$ the pullback of $\mathcal{F}$ to X_i. Assume that $\mathcal{F}_i$ is flat over T_i in dimensions $\geq n$ for all i. Let $t \in T$. Choose an index i and a point $t_i \in T_i$ mapping to t. Consider the cartesian diagram

$$\begin{array}{ccc} X_{\mathrm{Spec}(\mathcal{O}_{T,t})} & \longleftarrow & X_{\mathrm{Spec}(\mathcal{O}_{T_i,t_i})} \\ \downarrow & & \downarrow \\ \mathrm{Spec}(\mathcal{O}_{T,t}) & \longleftarrow & \mathrm{Spec}(\mathcal{O}_{T_i,t_i}) \end{array}$$

As the lower horizontal morphism is flat we see from More on Morphisms, Lemma 047C that the set $Z_i \subset X_{t_i}$ where $\mathcal{F}_i$ is not flat over T_i and the set $Z \subset X_t$ where $\mathcal{F}_T$ is not flat over T are related by the rule $Z_i = Z_{\kappa(t_i)}$. Hence we see that $\mathcal{F}_T$ is flat over T in dimensions $\geq n$ by Morphisms, Lemma 02FY.

Assume that f is quasi-compact and locally of finite presentation and that $\mathcal{F}$ is of finite presentation. In this paragraph we first reduce the proof of (2) to the case where f is of finite presentation. Let $T = \lim_{i \in I} T_i$ be a directed limit of affine S-schemes and assume that $\mathcal{F}_T$ is flat in dimensions $\geq n$. Set $X_i = X_{T_i} = X \times_S T_i$ and denote $\mathcal{F}_i$ the pullback of $\mathcal{F}$ to X_i. We have to show that $\mathcal{F}_i$ is flat in dimensions $\geq n$ for some i. Pick $i_0 \in I$ and replace I by $\{i \mid i \geq i_0\}$. Since T_{i_0} is affine (hence quasi-compact) there exist finitely many affine opens $W_j \subset S$, $j = 1, \ldots, m$ and an affine open overing $T_{i_0} = \bigcup_{j=1,\ldots,m} V_{j,i_0}$ such that $T_{i_0} \to S$ maps V_{j,i_0} into W_j. For $i \geq i_0$ denote $V_{j,i}$ the inverse image of V_{j,i_0} in T_i. If we can show, for each j, that there exists an i such that $\mathcal{F}_{V_{j,i_0}}$ is flat in dimensions $\geq n$, then we win. In this way we reduce to the case that S is affine. In this case X is quasi-compact and we can choose a finite affine open covering $X = W_1 \cup \ldots \cup W_m$. In this case the result for $(X \to S, \mathcal{F})$ is equivalent to the result for $(\coprod W_j, \coprod \mathcal{F}|_{W_j})$. Hence we may assume that f is of finite presentation.

Assume f is of finite presentation and $\mathcal{F}$ is of finite presentation. Let $U \subset X_T$ denote the open subscheme of points where $\mathcal{F}_T$ is flat over T, see More on Morphisms, Theorem 0399. By assumption the dimension of every fibre of $Z = X_T \setminus U$ over T has dimension $< n$. By Limits, Lemma 05M6 we can find a closed subscheme $Z \subset Z' \subset X_T$ such that $\dim(Z'_t) < n$ for all $t \in T$ and such that $Z' \to X_T$ is of finite presentation. By Limits, Lemmas 01ZM and 01ZP there exists an $i \in I$ and a closed subscheme $Z'_i \subset X_i$ of finite presentation whose base change to T is Z'. By Limits, Lemma 05M5 we may assume all fibres of $Z'_i \to T_i$ have dimension $< n$. By Limits, Lemma 05LY we may assume that $\mathcal{F}_i|_{X_i \setminus T'_i}$ is flat over T_i. This implies that $\mathcal{F}_i$ is flat in dimensions $\geq n$; here we use that $Z' \to X_T$ is of finite presentation, and hence the complement $X_T \setminus Z'$ is quasi-compact! Thus part (2) is proved and the proof of the lemma is complete. $\square$